Laboratorio 10: Implement intelligent search with full-text, vector, and hybrid queries

En este laboratorio se implementan y comparan tres enfoques de búsqueda inteligente en **Azure SQL Database** sobre la tabla `dbo.ProductReview` (140 reseñas de productos de AdventureWorksLT):

1. **Full-Text Search:** Búsqueda léxica mediante catálogos e índices de texto completo (`CONTAINS`, `FREETEXT`, `CONTAINSTABLE`, `FREETEXTTABLE`).
2. **Vector Search:** Búsqueda de similitud semántica exacta (`VECTOR_DISTANCE`) y aproximada (`VECTOR_SEARCH` con índice **DiskANN**) utilizando embeddings de 1536 dimensiones (`text-embedding-3-small`).
3. **Hybrid Search (RRF):** Combinación de búsqueda léxica (BM25) y semántica (ANN) mediante **Reciprocal Rank Fusion (RRF)**.

Arquitectura y Requisitos Previos

- **Base de datos:** Azure SQL Database (`AdventureWorksLT`, nivel Hyperscale Serverless).
- **Modelos en Microsoft Foundry:** `text-embedding-3-small` y `gpt-5.4-mini`.
- **Seguridad:** Identidad administrada asignada por el sistema (System-assigned Managed Identity) con el rol *Cognitive Services OpenAI User*.

Paso 1: Creación de la tabla ProductReview

Se ejecuta el script `product-reviews-insert.sql` y se verifica la carga de las 140 reseñas:

```
SELECT COUNT(*) AS TotalReviews FROM dbo.ProductReview;
GO
```



	TotalReviews
1	140

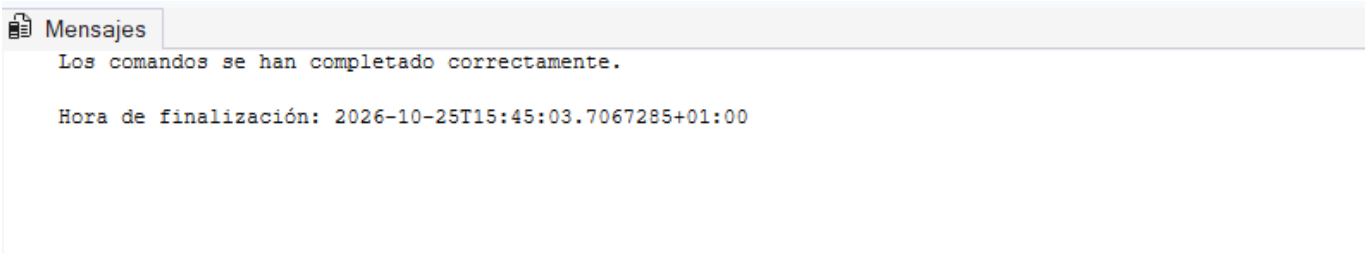
Paso 2: Credencial de base de datos y Modelo Externo (EXTERNAL MODEL)

Se configura la clave maestra, la credencial basada en `Managed Identity` y el objeto `EXTERNAL MODEL` apuntando al despliegue de `text-embedding-3-small`:

```
-- Create a database master key if one doesn't exist
IF NOT EXISTS (SELECT * FROM sys.symmetric_keys WHERE name =
'##MS_DatabaseMasterKey##')
    CREATE MASTER KEY ENCRYPTION BY PASSWORD = 'TuPasswordSeguro123!';
GO

-- Create a credential using Managed Identity
CREATE DATABASE SCOPED CREDENTIAL [https://proj-sqlailab-
resource.openai.azure.com]
WITH IDENTITY = 'Managed Identity', SECRET =
'{"resourceid":"https://cognitiveservices.azure.com"}';
GO

-- Create an external model reference for embeddings
CREATE EXTERNAL MODEL my_embedding_model
WITH (
    LOCATION = 'https://proj-sqlailab-
resource.openai.azure.com/openai/deployments/text-embedding-3-small/embeddings?
api-version=2024-10-21',
    API_FORMAT = 'Azure OpenAI',
    MODEL_TYPE = EMBEDDINGS,
    MODEL = 'text-embedding-3-small',
    CREDENTIAL = [https://proj-sqlailab-resource.openai.azure.com]
);
GO
```



Paso 3: Columna vectorial, generación de Embeddings e índice DiskANN

Se añade la columna `ReviewVector VECTOR(1536)`, se generan los embeddings en lotes de 30 registros controlando límites de tasa y se crea el índice vectorial **DiskANN**:

```
ALTER TABLE dbo.ProductReview
ADD ReviewVector VECTOR(1536);
GO

-- Generate embeddings in batches to avoid API rate limits
DECLARE @batchSize INT = 30;
DECLARE @rowsUpdated INT = 1;
DECLARE @retryCount INT;
DECLARE @maxRetries INT = 3;
```

```
WHILE @rowsUpdated > 0
BEGIN
    SET @retryCount = 0;

    RETRY:
    BEGIN TRY
        UPDATE TOP (@batchSize) r
        SET r.ReviewVector = AI_GENERATE_EMBEDDINGS(
            p.Name + ' - ' + r.ReviewTitle + ': ' + r.ReviewText
            USE MODEL my_embedding_model)
        FROM dbo.ProductReview r
        INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
        WHERE r.ReviewVector IS NULL;

        SET @rowsUpdated = @@ROWCOUNT;

        IF @rowsUpdated > 0
            WAITFOR DELAY '00:00:02';
    END TRY
    BEGIN CATCH
        SET @retryCount += 1;
        IF @retryCount <= @maxRetries
        BEGIN
            PRINT 'Rate limited. Retrying in 5 seconds... (Attempt '
                + CAST(@retryCount AS NVARCHAR(10)) + ' of '
                + CAST(@maxRetries AS NVARCHAR(10)) + ')';
            WAITFOR DELAY '00:00:05';
            GOTO RETRY;
        END
        ELSE
            THROW;
    END CATCH
END
GO

-- Check how many reviews have embeddings
SELECT
    COUNT(*) AS TotalReviews,
    COUNT(ReviewVector) AS ReviewsWithEmbeddings
FROM dbo.ProductReview;
GO

-- Enable preview features and create DiskANN vector index
ALTER DATABASE SCOPED CONFIGURATION SET PREVIEW_FEATURES = ON;
GO
ALTER DATABASE SCOPED CONFIGURATION SET ALLOW_STALE_VECTOR_INDEX = ON;
GO

CREATE VECTOR INDEX IX_Review_ReviewVector
ON dbo.ProductReview(ReviewVector)
WITH (METRIC = 'cosine', TYPE = 'DISKANN');
GO
```


Resultados Mensajes

	TotalReviews	ReviewsWithEmbeddings
1	140	140

Paso 4: Creación del Catálogo e Índice Full-Text

Se crea el catálogo por defecto y el índice de texto completo sobre `ReviewTitle` y `ReviewText` con seguimiento de cambios automático:

```
CREATE FULLTEXT CATALOG ProductReviewCatalog AS DEFAULT;
GO

CREATE FULLTEXT INDEX ON dbo.ProductReview
(
    ReviewTitle LANGUAGE 1033,
    ReviewText LANGUAGE 1033
)
KEY INDEX PK_ProductReview
ON ProductReviewCatalog
WITH (CHANGE_TRACKING AUTO);
GO

-- Check full-text index status
SELECT
    OBJECTPROPERTY(OBJECT_ID('dbo.ProductReview'), 'TableFullTextPopulateStatus')
AS PopulateStatus,
    OBJECTPROPERTY(OBJECT_ID('dbo.ProductReview'), 'TableHasActiveFulltextIndex')
AS HasActiveIndex;
GO
```


Resultados Mensajes

	PopulateStatus	HasActiveIndex
1	1	1

Paso 5: Búsqueda con Predicados Full-Text

Pruebas de coincidencia exacta (`CONTAINS`), formas inflexivas (`FREETEXT`), ranking de relevancia (`CONTAINSTABLE`) y búsqueda por prefijos (`"comfort*"`):

```
-- 1. Exact word match with CONTAINS
SELECT
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    p.Name AS ProductName
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
WHERE CONTAINS(r.ReviewText, 'puncture');
GO

-- 2. Inflectional phrase search with FREETEXT
SELECT TOP 10
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    p.Name AS ProductName
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
WHERE FREETEXT(r.ReviewText, 'warm gloves for cold winter commuting');
GO

-- 3. Ranked results with CONTAINSTABLE and FORMSOF
SELECT TOP 10
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    p.Name AS ProductName,
    ft.[RANK] AS FullTextRank
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
INNER JOIN CONTAINSTABLE(dbo.ProductReview, (ReviewTitle, ReviewText),
    'FORMSOF(INFLECTIONAL, mountain) AND FORMSOF(INFLECTIONAL, trail)') AS ft
    ON r.ReviewID = ft.[KEY]
ORDER BY ft.[RANK] DESC;
GO

-- 4. Prefix search
SELECT
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    p.Name AS ProductName
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
WHERE CONTAINS(r.ReviewText, '"comfort*"');
GO
```



Resultados Mensajes

	ReviewTitle	ReviewText	Rating	ProductName
1	Extraordinary tire life and puncture resistance	Got 4,500 miles out of a pair of HL Road Tires wi...	5	ML Road Frame - Red, 58
2	Puncture magnet	The LL Road Tire seems to attract every piece of...	2	LL Road Frame - Black, 52
3	Saved me 30 miles from anywhere	The patch kit fixed a sidewall puncture 30 miles f...	4	LL Mountain Frame - Black, 44

	ReviewTitle	ReviewText	Rating	ProductName
1	Perfect commuter road bike	I ride 12 miles each way to work and the Road-150 h...	5	ML Fork
2	Solid daily commuter	Switched from the bus to cycling 9 miles to my office. ...	4	ML Road Frame-W - Yellow, 48
3	Reliable rain or shine commuter	Replaced my car with the Road-650 Red for a 7-mile ...	4	ML Mountain Frame - Black, 40
4	Best budget road bike for daily riding	Compared the Road-650 Red against three competit...	5	ML Mountain Frame - Black, 40
5	Visible and ventilated for night riding	The Sport-100 in black has reflective accents that ligh...	5	Mountain Bike Socks, M
6	Okay in winter but slips on ice	The LL Mountain Tire is serviceable on cold hard-pac...	3	HL Mountain Frame - Black, 42
7	Best all-weather road tire I have us...	The HL Road Tire performs equally well in rain, heat, ...	5	ML Road Frame - Red, 58
8	Does the job but nothing special	The Road Tire Tube holds air adequately but loses a ...	3	Touring Tire Tube

	ReviewTitle	ReviewText	Rating	ProductName	FullTextRank
1	Durability after years of hard riding	Three years and 8,000 miles of aggressive trail ridi...	5	Mountain-100 Black, 38	0
2	Rear shock leaked oil after five months	The Mountain-100 Black frame is outstanding but ...	2	Mountain-100 Black, 38	0
3	Great mid-range trail bike	The Mountain-200 Black handles Pacific Northwe...	4	Mountain-200 Black, 42	0
4	Fork bottoms out on bigger hits	The Mountain-200 Black is solid for moderate trail...	3	Mountain-200 Black, 42	0
5	Massive upgrade from my old hardtail	Replaced a 14-year-old hardtail with the Mountain...	5	Mountain-200 Silver, 38	0
6	Solid entry into serious trail riding	Upgraded from a department store bike to the Mo...	4	Mountain-300 Black, 44	0
7	Frame integrity after years of hard use	Riding the Mountain-300 Black for over two years ...	4	Mountain-300 Black, 44	0
8	Perfect weekend trail companion	We take the Mountain-500 Silver out every Saturd...	5	Road-250 Black, 52	0

	ReviewTitle	ReviewText	Rating	ProductName
1	Sunday ride perfection	The Road-550-W is my go-to for Sunday morning ri...	5	HL Mountain Rear Wheel
2	Handlebars flex during hard efforts	Comfortable for cruising, but when I stand and sprint...	3	LL Road Rear Wheel
3	Great first road bike	Transitioned from a hybrid to the Road-650 Black a...	4	ML Road Frame-W - Ye...
4	Proportions that actually make se...	The Road-350-W has the right proportions for small...	5	ML Mountain Front Wh...
5	Fit that works for smaller riders	The Mountain-400-W solved every fit issue I had wit...	4	Road-250 Black, 44
6	Great family weekend bike	Bought the Mountain-500 Black for Saturday rides w...	4	Road-550-W Yellow, 44

Paso 6: Búsqueda por Similitud Vectorial (VECTOR_DISTANCE y VECTOR_SEARCH)

Comparación entre búsqueda exacta KNN (`VECTOR_DISTANCE`) y búsqueda aproximada ANN sobre el grafo DiskANN (`VECTOR_SEARCH`), además de inspección de metadatos con `VECTORPROPERTY`:

```
-- 1. Exact Nearest Neighbor (KNN) with VECTOR_DISTANCE
DECLARE @searchText NVARCHAR(1000) = 'comfortable seat for long distance touring';
DECLARE @searchVector VECTOR(1536);

SELECT @searchVector = AI_GENERATE_EMBEDDINGS(@searchText USE MODEL
my_embedding_model);

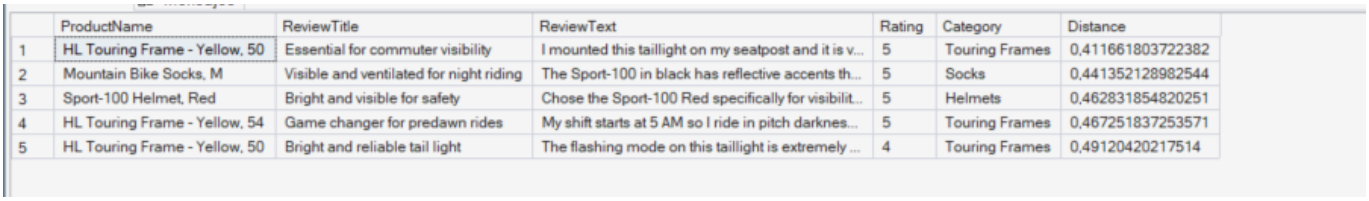
SELECT TOP 5
    p.Name AS ProductName,
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    VECTOR_DISTANCE('cosine', @searchVector, r.ReviewVector) AS Distance
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
ORDER BY Distance;
GO

-- 2. Approximate Nearest Neighbor (ANN) with VECTOR_SEARCH (Sintaxis actualizada)
DECLARE @searchText NVARCHAR(1000) = 'something to keep me visible when riding at
night';
DECLARE @searchVector VECTOR(1536);

SELECT @searchVector = AI_GENERATE_EMBEDDINGS(@searchText USE MODEL
my_embedding_model);
```

```
SELECT TOP (5)
    p.Name AS ProductName,
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    pc.Name AS Category,
    vs.distance AS Distance
FROM VECTOR_SEARCH(
    TABLE = dbo.ProductReview AS r,
    COLUMN = ReviewVector,
    SIMILAR_TO = @searchVector,
    METRIC = 'cosine'
) AS vs
INNER JOIN SalesLT.Product p
    ON r.ProductID = p.ProductID
INNER JOIN SalesLT.ProductCategory pc
    ON p.ProductCategoryID = pc.ProductCategoryID
ORDER BY vs.distance ASC;
GO

-- 3. Inspect vector metadata
SELECT TOP 1
    VECTORPROPERTY(ReviewVector, 'Dimensions') AS VectorDimensions,
    VECTORPROPERTY(ReviewVector, 'BaseType') AS VectorBaseType
FROM dbo.ProductReview
WHERE ReviewVector IS NOT NULL;
GO
```



	ProductName	ReviewTitle	ReviewText	Rating	Category	Distance
1	HL Touring Frame - Yellow, 50	Essential for commuter visibility	I mounted this taillight on my seatpost and it is v...	5	Touring Frames	0,411661803722382
2	Mountain Bike Socks, M	Visible and ventilated for night riding	The Sport-100 in black has reflective accents th...	5	Socks	0,441352128982544
3	Sport-100 Helmet, Red	Bright and visible for safety	Chose the Sport-100 Red specifically for visibilit...	5	Helmets	0,462831854820251
4	HL Touring Frame - Yellow, 54	Game changer for predawn rides	My shift starts at 5 AM so I ride in pitch darknes...	5	Touring Frames	0,467251837253571
5	HL Touring Frame - Yellow, 50	Bright and reliable tail light	The flashing mode on this taillight is extremely ...	4	Touring Frames	0,49120420217514

Paso 7: Búsqueda Híbrida con Reciprocal Rank Fusion (RRF)

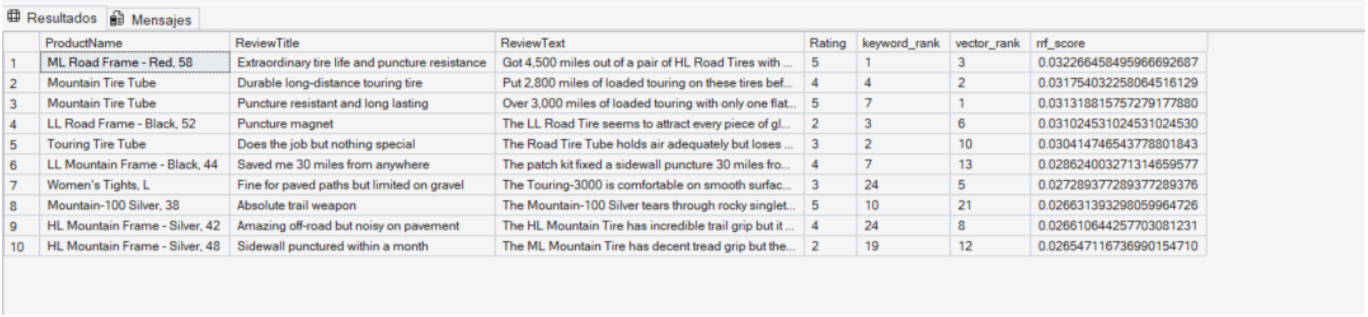
Fusión de rankings léxicos (`FREETEXTTABLE`) y vectoriales (`VECTOR_SEARCH`) mediante la fórmula RRF_Score = 1/(k + rank_keyword) + 1/(k + rank_vector) con k = 60:

```
DECLARE @searchText NVARCHAR(1000) = 'durable tires that resist punctures on rough
terrain';
DECLARE @searchVector VECTOR(1536);
DECLARE @topN INT = 50;
DECLARE @rrfK INT = 60;

SELECT @searchVector = AI_GENERATE_EMBEDDINGS(@searchText USE MODEL
my_embedding_model);

WITH keyword_search AS (
```

```
    SELECT TOP(@topN)
        r.ReviewID,
        RANK() OVER (ORDER BY ft.[RANK] DESC) AS keyword_rank
    FROM dbo.ProductReview r
    INNER JOIN FREETEXTTABLE(dbo.ProductReview, (ReviewTitle, ReviewText),
@searchText) AS ft
        ON r.ReviewID = ft.[KEY]
),
vector_search AS (
    SELECT TOP(@topN)
        ReviewID,
        RANK() OVER (ORDER BY distance ASC) AS vector_rank
    FROM (
        SELECT
            r.ReviewID,
            vs.distance
        FROM VECTOR_SEARCH(
            TABLE = dbo.ProductReview AS r,
            COLUMN = ReviewVector,
            SIMILAR_TO = @searchVector,
            METRIC = 'cosine'
        ) AS vs
    ) AS similar_reviews
),
combined AS (
    SELECT
        COALESCE(ks.ReviewID, vs.ReviewID) AS ReviewID,
        ks.keyword_rank,
        vs.vector_rank,
        COALESCE(1.0 / (@rrfK + ks.keyword_rank), 0.0) +
        COALESCE(1.0 / (@rrfK + vs.vector_rank), 0.0) AS rrf_score
    FROM keyword_search ks
    FULL OUTER JOIN vector_search vs ON ks.ReviewID = vs.ReviewID
)
SELECT TOP 10
    p.Name AS ProductName,
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    c.keyword_rank,
    c.vector_rank,
    c.rrf_score
FROM combined c
INNER JOIN dbo.ProductReview r ON c.ReviewID = r.ReviewID
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
ORDER BY c.rrf_score DESC;
GO
```



Resultados Mensajes

	ProductName	ReviewTitle	ReviewText	Rating	keyword_rank	vector_rank	rrf_score
1	ML Road Frame - Red, 58	Extraordinary tire life and puncture resistance	Got 4,500 miles out of a pair of HL Road Tires with ...	5	1	3	0.032266458495966692687
2	Mountain Tire Tube	Durable long-distance touring tire	Put 2,800 miles of loaded touring on these tires bef...	4	4	2	0.031754032258064516129
3	Mountain Tire Tube	Puncture resistant and long lasting	Over 3,000 miles of loaded touring with only one flat...	5	7	1	0.031318815757279177880
4	LL Road Frame - Black, 52	Puncture magnet	The LL Road Tire seems to attract every piece of gl...	2	3	6	0.031024531024531024530
5	Touring Tire Tube	Does the job but nothing special	The Road Tire Tube holds air adequately but loses ...	3	2	10	0.030414746543778801843
6	LL Mountain Frame - Black, 44	Saved me 30 miles from anywhere	The patch kit fixed a sidewall puncture 30 miles fro...	4	7	13	0.028624003271314659577
7	Women's Tights, L	Fine for paved paths but limited on gravel	The Touring-3000 is comfortable on smooth surfac...	3	24	5	0.027289377289377289376
8	Mountain-100 Silver, 38	Absolute trail weapon	The Mountain-100 Silver tears through rocky singlet...	5	10	21	0.026631393298059964726
9	HL Mountain Frame - Silver, 42	Amazing off-road but noisy on pavement	The HL Mountain Tire has incredible trail grip but it ...	4	24	8	0.026610644257703081231
10	HL Mountain Frame - Silver, 48	Sidewall punctured within a month	The ML Mountain Tire has decent tread grip but the...	2	19	12	0.026547116736990154710

Paso 8: Comparativa de los Tres Enfoques

Evaluación de la consulta `'comfortable bike for long weekend rides with the family'` bajo los tres métodos para contrastar resultados léxicos, semánticos e híbridos:

```
-- 1. Full-text search only
SELECT TOP 5
    p.Name AS ProductName,
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    ft.[RANK] AS FullTextRank
FROM dbo.ProductReview r
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
INNER JOIN FREETEXTTABLE(dbo.ProductReview, (ReviewTitle, ReviewText),
    'comfortable bike for long weekend rides with the family') AS ft
    ON r.ReviewID = ft.[KEY]
ORDER BY ft.[RANK] DESC;
GO

-- 2. Vector search only (Sintaxis actualizada)
DECLARE @searchText NVARCHAR(1000) = 'comfortable bike for long weekend rides with
the family';
DECLARE @searchVector VECTOR(1536);

SELECT @searchVector = AI_GENERATE_EMBEDDINGS(@searchText USE MODEL
my_embedding_model);

SELECT TOP (5)
    p.Name AS ProductName,
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    vs.distance AS Distance
FROM VECTOR_SEARCH(
    TABLE = dbo.ProductReview AS r,
    COLUMN = ReviewVector,
    SIMILAR_TO = @searchVector,
    METRIC = 'cosine'
) AS vs
INNER JOIN SalesLT.Product p
    ON r.ProductID = p.ProductID
```

```
ORDER BY vs.distance ASC;
GO

-- 3. Hybrid search with RRF (Sintaxis actualizada)
DECLARE @searchText NVARCHAR(1000) = 'comfortable bike for long weekend rides with
the family';
DECLARE @searchVector VECTOR(1536);
DECLARE @topN INT = 50;
DECLARE @rrfK INT = 60;

SELECT @searchVector = AI_GENERATE_EMBEDDINGS(@searchText USE MODEL
my_embedding_model);

WITH keyword_search AS (
    SELECT TOP(@topN)
        r.ReviewID,
        RANK() OVER (ORDER BY ft.[RANK] DESC) AS keyword_rank
    FROM dbo.ProductReview r
    INNER JOIN FREETEXTTABLE(dbo.ProductReview, (ReviewTitle, ReviewText),
@searchText) AS ft
        ON r.ReviewID = ft.[KEY]
),
vector_search AS (
    SELECT TOP(@topN)
        ReviewID,
        RANK() OVER (ORDER BY distance ASC) AS vector_rank
    FROM (
        SELECT
            r.ReviewID,
            vs.distance
        FROM VECTOR_SEARCH(
            TABLE = dbo.ProductReview AS r,
            COLUMN = ReviewVector,
            SIMILAR_TO = @searchVector,
            METRIC = 'cosine'
        ) AS vs
    ) AS similar_reviews
),
combined AS (
    SELECT
        COALESCE(ks.ReviewID, vs.ReviewID) AS ReviewID,
        ks.keyword_rank,
        vs.vector_rank,
        COALESCE(1.0 / (@rrfK + ks.keyword_rank), 0.0) +
        COALESCE(1.0 / (@rrfK + vs.vector_rank), 0.0) AS rrf_score
    FROM keyword_search ks
    FULL OUTER JOIN vector_search vs ON ks.ReviewID = vs.ReviewID
)
SELECT TOP 5
    p.Name AS ProductName,
    r.ReviewTitle,
    r.ReviewText,
    r.Rating,
    c.keyword_rank,
```

```
    c.vector_rank,
    c.rrf_score
FROM combined c
INNER JOIN dbo.ProductReview r ON c.ReviewID = r.ReviewID
INNER JOIN SalesLT.Product p ON r.ProductID = p.ProductID
ORDER BY c.rrf_score DESC;
GO
```

Resultados Mensajes

	ProductName	ReviewTitle	ReviewText	Rating	FullTextRank
1	Water Bottle - 30 oz.	Stays cold for hours on long rides	We do 3-hour family rides along the river trail every...	5	108
2	Water Bottle - 30 oz.	Perfect family hydration solution	Bought four of these 30 oz water bottles – one for ...	5	108
3	Long-Sleeve Logo Jersey, L	Matching jerseys for the whole family	Ordered the Long-Sleeve Logo Jersey in different ...	5	92
4	Road-550-W Yellow, 44	Great family weekend bike	Bought the Mountain-500 Black for Saturday rides ...	4	88
5	ML Road Frame-W - Yellow, 48	Great first road bike	Transitioned from a hybrid to the Road-650 Black ...	4	76

	ProductName	ReviewTitle	ReviewText	Rating	Distance
1	Road-550-W Yellow, 44	Great family weekend bike	Bought the Mountain-500 Black for Saturday rides w...	4	0,349750995635986
2	Road-250 Black, 52	Perfect weekend trail companion	We take the Mountain-500 Silver out every Saturday ...	5	0,464610040187836
3	HL Mountain Rear Wheel	Sunday ride perfection	The Road-550-W is my go-to for Sunday morning ri...	5	0,468393623828888
4	Men's Sports Shorts, L	Capable mid-range touring platform	The Touring-2000 handled a 400-mile charity ride w...	4	0,475968539714813
5	Touring-Panniers, Large	Built for the long haul	Took the Touring-1000 Yellow on a 600-mile ride al...	5	0,495013236999512

	ProductName	ReviewTitle	ReviewText	Rating	keyword_rank	vector_rank	rrf_score
1	Road-550-W Yellow, 44	Great family weekend bike	Bought the Mountain-500 Black for Saturday rides ...	4	4	1	0.032018442622950819672
2	Water Bottle - 30 oz.	Stays cold for hours on long rides	We do 3-hour family rides along the river trail every...	5	1	7	0.031318815757279177880
3	Water Bottle - 30 oz.	Perfect family hydration solution	Bought four of these 30 oz water bottles – one for ...	5	1	21	0.028739121635296498684
4	Men's Bib-Shorts, S	Touring made easy for a first-timer	Never toured before but bought the Touring-3000 ...	5	11	9	0.028577260665441926923
5	Road-250 Black, 52	Perfect weekend trail companion	We take the Mountain-500 Silver out every Saturda...	5	21	2	0.028474711270410195141